Gabriel Eze

chieloka.eze@centre.edu | +1 (859) 329-8224 | https://www.linkedin.com/in/gabriel-eze/

github.com/ezegabriel | https://ezegabriel.github.io/www/

EDUCATION

Bachelor of Science in Data Science and Mathematics, *Centre College* 08/22 – 05/26 | GPA: 3.55
Relevant coursework: **Abstract Algebra, Advanced Interpolation Methods, Applied Machine Learning, Bayesian Data Analysis, Building AI-Powered Applications, Data Manipulation, Data Visualization, How to Lie With Statistics, Linear Algebra I & II, Mathematical Statistics, Modern Calculus II & III, Probability Theory, Python Programming and Problem Solving, Real Analysis, Statistical Modeling.**

SKILLS

Computing
Combinatorics, Harmonic Analysis, Q-Ball Imaging

Programming
MATLAB, Python, R, SQL

LLM
Chunking, Doc2Query, Embeddings, Hit@K, Prompt Engineering, Red-Team Testing, Semantic Search, Tool Use

Tools
ChromaDB, Git, HTML/CSS, JupyterLab, LaTeX, MySQL, Pydantic, Quarto, Regex, Tableau, VS Code, YAML

RESEARCH

Stack Domination Density of Hypercubes, *Graph Theory* 05/25 – 08/25 | Dr. Brauch, Dr. Wiglesworth
- Formulated and proved results for stack domination density in $\boldsymbol{C}m \times \boldsymbol{P}2 \times \boldsymbol{P}n$, deriving bounds across residue classes of m.
- Designed modular "Clockwise" and "Domination" algorithms to construct minimal dominating sets and validate proofs through large-scale computation.
- Co-authored a 38-page research manuscript integrating combinatorial proofs and algorithmic verification for potential undergraduate publication.

Pitch Modeling, *Carnegie Mellon SURE Camp* 06/24 – 07/24 | Quang Nguyen, Dr. Yurko
- Conducted advanced statistical analysis and modeling to predict pitch effectiveness (Stuff+) for MLB players, leveraging large datasets from FanGraphs and Baseball Savant.
- Applied Random Forest, Lasso regression, and HGBR models to analyze the complete arsenal of pitches per pitcher, enhancing predictive accuracy by modeling multiple pitch types and providing competitive RMSE results.
- Developed a pitch recommendation tool that provided actionable insights for MLB teams, optimizing player performance by suggesting ideal pitch sequencing based on past game data.

Sports Analytics 05/23 – 07/23 | Dr. Heath
- Utilized Python to lead data cleaning and parsing efforts within a complex big data pipeline, ensuring data accuracy and reliability for basketball game analysis.
- Tracked possession data and employed advanced statistical techniques to assess player substitutions, providing vital insights for informed coaching decisions and game outcome analysis.
- Collaborated on the development of win probability models based on real-time game data, empowering the coaching staff with actionable insights for strategic game planning.

PERSONAL PROJECTS

DNA Sequence Analysis 10/22 – 12/22
- Developed a Python program for precise comparison of Short Tandem Repeat (STR) repetitions within DNA sequences to a database, facilitating accurate DNA matching.
- Designed and implemented robust data validation techniques to ensure the reliability of identified DNA sequences, including the elimination of anomalies and detection of zero repetitions.
- Created a user-friendly interface for inputting files and provided real-time matching results by comparing repetitive DNA strings with data stored in a CSV file, showcasing proficiency in Python and data analysis.

Web Scraper 05/23 – 07/23
- Developed a Python-based web scraper capable of efficiently extracting targeted data from websites. Employed libraries such as requests and BeautifulSoup for web requests, HTML parsing, and data extraction.
- Demonstrated proficiency in data parsing and manipulation, enabling the scraper to retrieve valuable information, such as news articles or product details, and store it in structured formats like CSV files or databases.
- Enhanced user interaction by creating an intuitive command-line interface for inputting website URLs and data extraction parameters. The project highlighted skills in web scraping, data extraction, and command-line interface design.